

EXHIBIT 1

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U.S. Patent No. 9,865,546 Infringement Chart

Claim	Tower '546 Accused Process
[1pre] A method for forming a contact in a dielectric layer to a semiconductor substrate, the method comprising:	To the extent the preamble is limiting, Tower's SiGe, BiCMOS, and RF processes (including its 65nm RF process) (hereinafter, "Tower '546 Accused Process") comprise a method for forming a contact in a dielectric layer to a semiconductor substrate, as explained further in the limitations below.¹ For example, the Tower '546 Accused Process is used to manufacture the Qorvo QM81026 chip. The Tower '546 Accused Process comprises forming a contact in a dielectric layer to a semiconductor substrate, as shown in the reverse engineering analysis below, which shows a cross-section of the Qorvo QM81026 chip. 

¹ GlobalFoundries reserves the right to identify to identify different or additional Tower '546 Accused Processes and amend or supplement these charts after further investigation and discovery, claim construction, or other meriting circumstances. Such supplemental information may include, but is not limited to, source code, data sheets, design specifications, deposition testimony, testing information, reference designs implementation and utilization information, and/or schematics.

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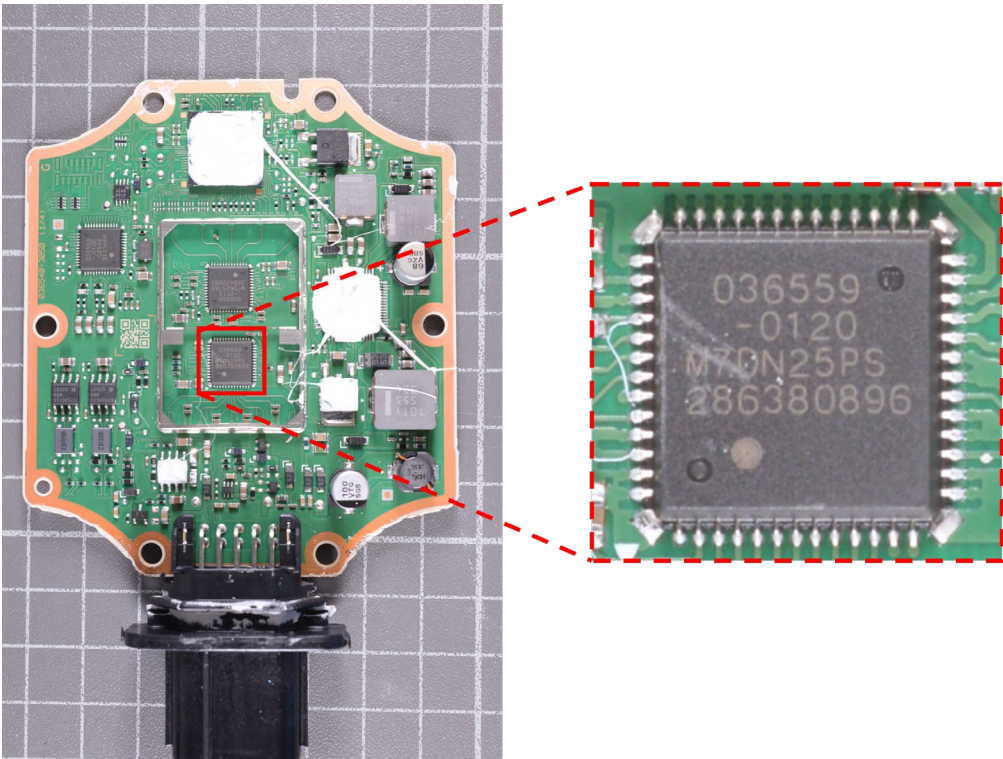
Claim	Tower '546 Accused Process
	For example, the Toyota Blind Spot Monitor Sensor (#88162-0R031) includes a module manufactured by DENSO Corporation (identified as DNSRR004), which further includes a die manufactured by Tower (identified as M7DN25PS) that is, on information and belief, representative of devices manufactured by Tower that rely on its SBC18 process.  GF's contentions are based on information reasonably available at this stage of the case and GF's investigation to date. GF reserves the right to rely on reverse engineering and/or other testing, which will further confirm the Tower '546 Accused Process comprises a method for forming a contact in a dielectric layer to a semiconductor substrate.

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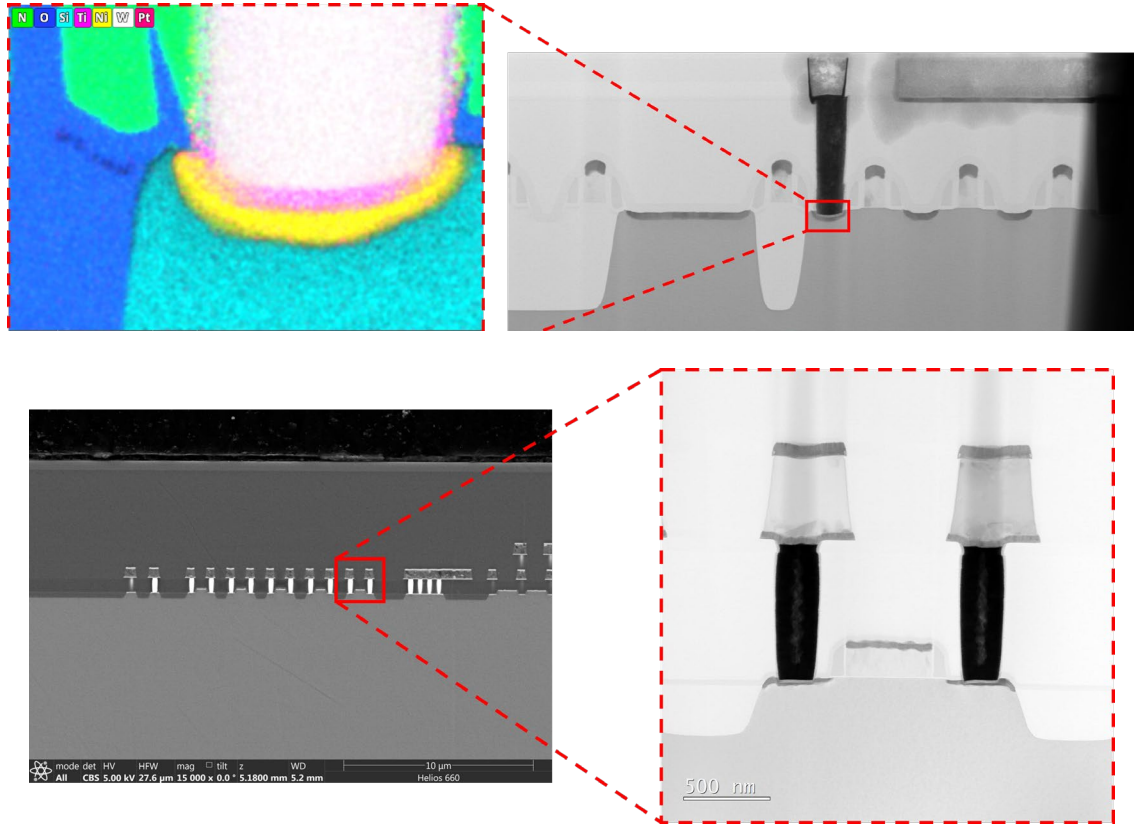
Claim	Tower '546 Accused Process
	To the extent this limitation is found not to be literally present, this limitation is present under the doctrine of equivalents because the Tower '546 Accused Process performs substantially the same function, in substantially the same way, to achieve substantially the same result as the claimed limitation, and any differences are insubstantial. To date, Respondents have not alleged that this limitation is not literally present in the Tower '546 Accused Process.
[1a] forming a contact opening in the dielectric layer over the semiconductor substrate to expose an upper portion of the semiconductor substrate;	The Tower '546 Accused Process comprises forming a contact opening in the dielectric layer over the semiconductor substrate to expose an upper portion of the semiconductor substrate. 

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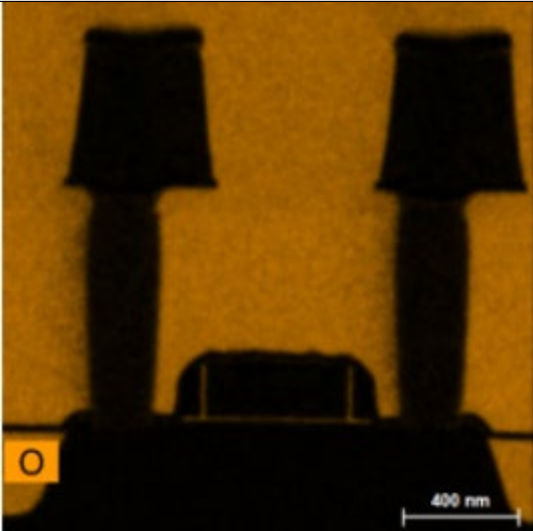
Claim	Tower '546 Accused Process
	 GF's contentions are based on information reasonably available at this stage of the case and GF's investigation to date. GF reserves the right to rely on reverse engineering and/or other testing, which will further confirm the Tower '546 Accused Process comprises forming a contact opening in the dielectric layer over the semiconductor substrate to expose an upper portion of the semiconductor substrate. To the extent this limitation is found not to be literally present, this limitation is present under the doctrine of equivalents because the Tower '546 Accused Process performs substantially the same function, in substantially the same way, to achieve substantially the same result as the claimed limitation, and any differences are insubstantial. To date, Respondents have not alleged that this limitation is not literally present in the Tower '546 Accused Process.
[1b] depositing a first liner layer to conformally coat the contact opening;	The Tower '546 Accused Process comprises depositing a first liner layer to conformally coat the contact opening.

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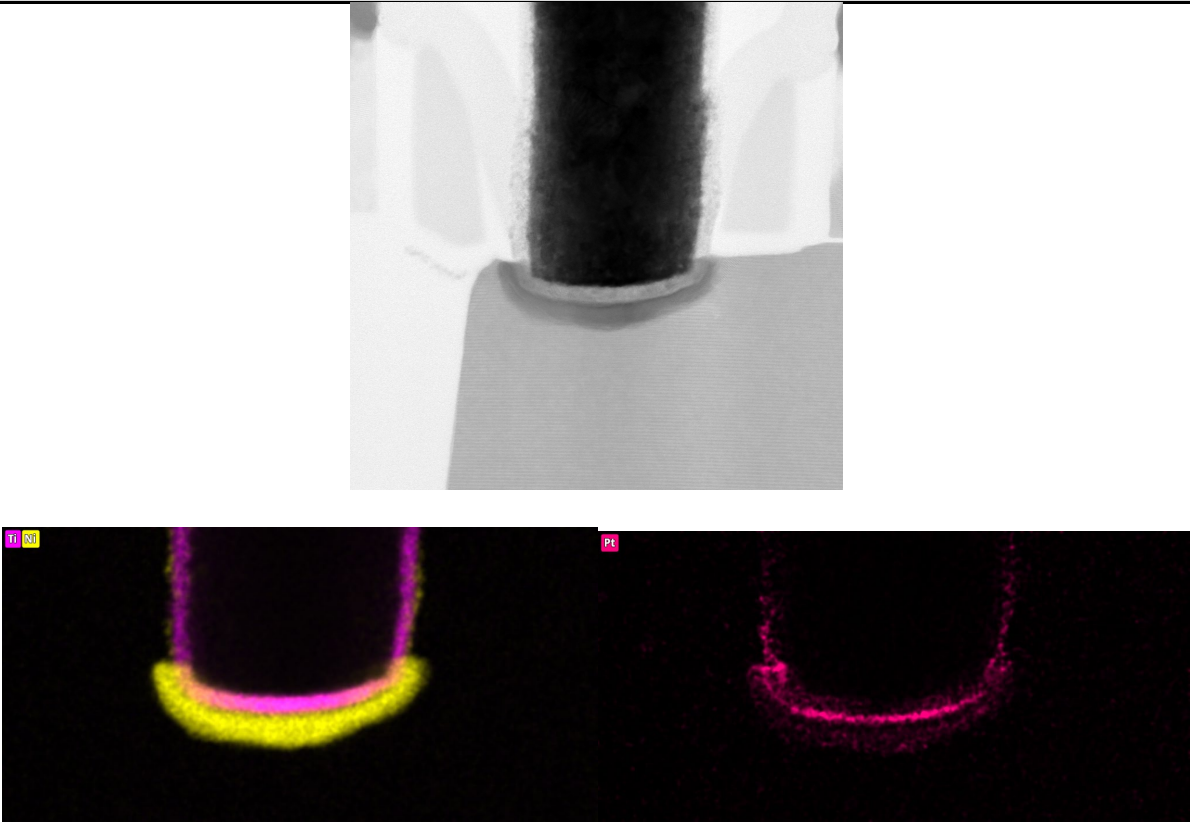
Claim	Tower '546 Accused Process
	

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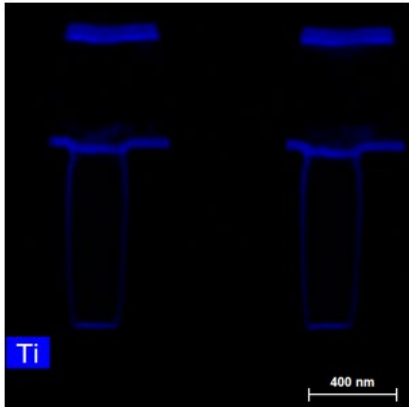
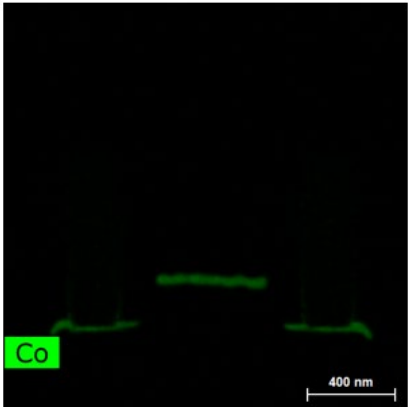
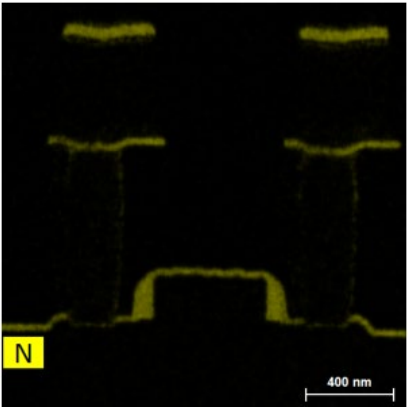
Claim	Tower '546 Accused Process
	  Ti  Co  N GF's contentions are based on information reasonably available at this stage of the case and GF's investigation to date. GF reserves the right to rely on reverse engineering and/or other testing, which will further confirm the Tower '546 Accused Process comprises depositing a first liner layer to conformally coat the contact opening.

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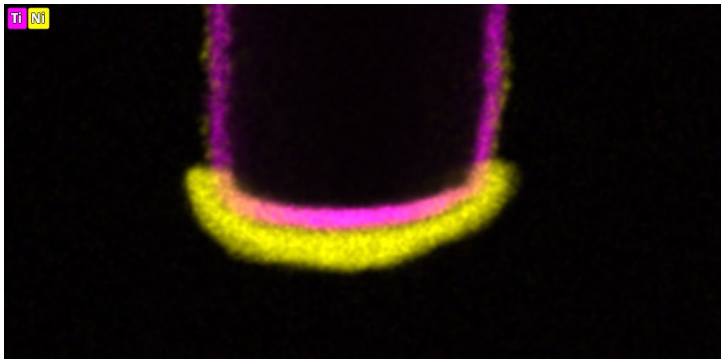
Claim	Tower '546 Accused Process
	To the extent this limitation is found not to be literally present, this limitation is present under the doctrine of equivalents because the Tower '546 Accused Process performs substantially the same function, in substantially the same way, to achieve substantially the same result as the claimed limitation, and any differences are insubstantial. To date, Respondents have not alleged that this limitation is not literally present in the Tower '546 Accused Process.
[1c] causing a portion of the first liner layer to diffuse into the upper portion of the semiconductor substrate to form a first intermix region at the upper portion of the semiconductor substrate;	The Tower '546 Accused Process comprises causing a portion of the first liner layer to diffuse into the upper portion of the semiconductor substrate to form a first intermix region at the upper portion of the semiconductor substrate. For example, the Tower '546 Accused Process includes causing a portion of the first liner layer to diffuse into the substrate, thereby forming a first intermix region. 

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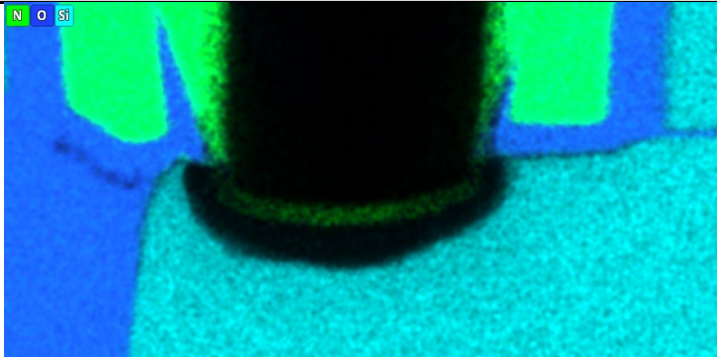
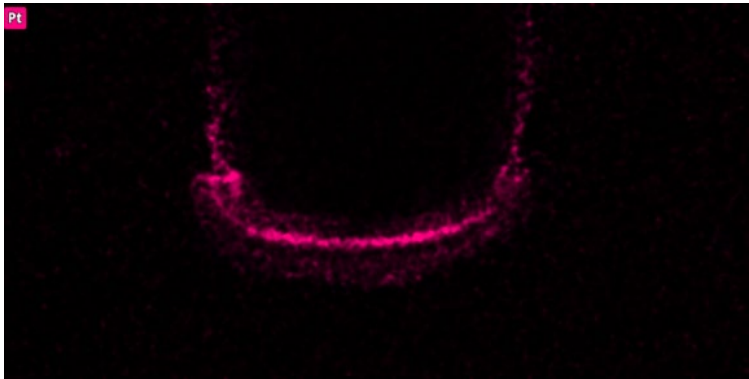
Claim	Tower '546 Accused Process
	  The arrow in the reverse engineering analysis below corresponds to the arrow at the top of the ensuing material analysis chart (i.e., such that the upward direction in the reverse engineering analysis below corresponds to the right-to-left direction in the ensuing material analysis chart).

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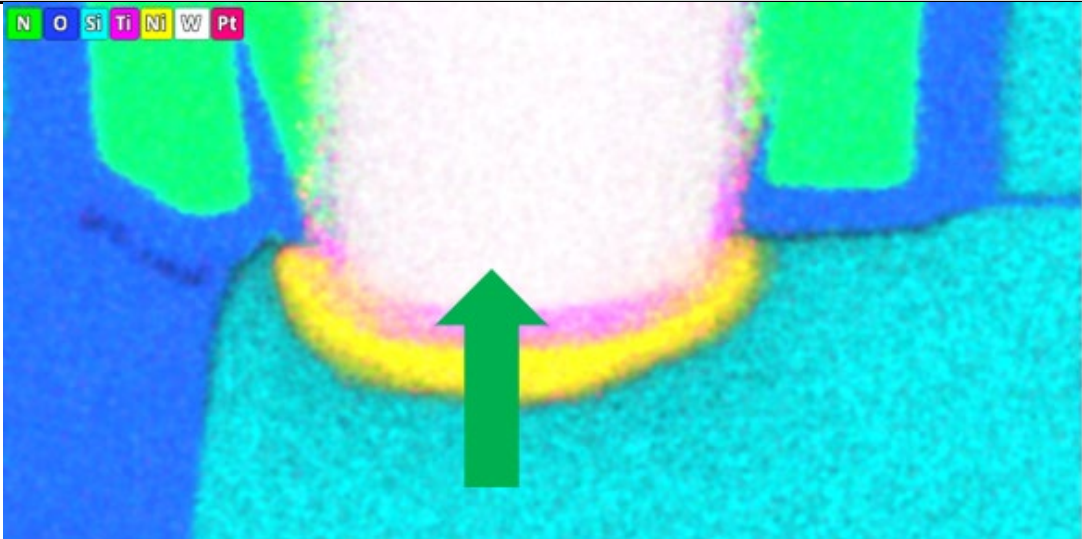
Claim	Tower '546 Accused Process
	A material analysis chart (likely an EDS line scan) showing the composition of a device cross-section. The x-axis represents the spatial position, and the y-axis represents the intensity of various elements. The elements listed at the top are N (Nitrogen), O (Oxygen), Si (Silicon), Ti (Titanium), Ni (Nickel), W (Tungsten), and Pt (Platinum). The chart shows a blue region on the left, a green region in the middle, and a red region on the right. A yellow arrow points to a specific region in the green area, indicating an intermix region where the first liner layer has diffused into the silicon substrate. As shown in the material analysis chart below, the intermix region created by the Tower '546 Accused Process includes a portion of the first liner layer diffusing into the silicon substrate.

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Claim	Tower '546 Accused Process																																																																																								
	The graph displays the weight fraction (%) of several elements as a function of position (nm). The x-axis ranges from 0 to 50 nm, and the y-axis ranges from 0 to 80%. A large green arrow points from right to left above the graph, indicating the direction of diffusion. The graph shows a sharp drop in W (white line) from ~80% at 0 nm to ~10% at 10 nm. Ti (magenta) and Ni (yellow) show peaks around 15-20 nm. Si (cyan) rises sharply after 30 nm. N (green), O (blue), and Pt (pink) remain at low levels. <table><tr><th>Position (nm)</th><th>N (%)</th><th>O (%)</th><th>Si (%)</th><th>Ti (%)</th><th>Ni (%)</th><th>W (%)</th><th>Pt (%)</th></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>80</td><td>0</td></tr><tr><td>10</td><td>0</td><td>0</td><td>0</td><td>5</td><td>5</td><td>10</td><td>0</td></tr><tr><td>15</td><td>0</td><td>0</td><td>0</td><td>45</td><td>15</td><td>10</td><td>0</td></tr><tr><td>20</td><td>0</td><td>0</td><td>0</td><td>20</td><td>45</td><td>10</td><td>0</td></tr><tr><td>25</td><td>0</td><td>0</td><td>0</td><td>10</td><td>45</td><td>10</td><td>0</td></tr><tr><td>30</td><td>0</td><td>0</td><td>0</td><td>10</td><td>45</td><td>10</td><td>0</td></tr><tr><td>35</td><td>0</td><td>0</td><td>0</td><td>10</td><td>45</td><td>10</td><td>0</td></tr><tr><td>40</td><td>0</td><td>0</td><td>80</td><td>10</td><td>45</td><td>10</td><td>0</td></tr><tr><td>45</td><td>0</td><td>0</td><td>80</td><td>10</td><td>45</td><td>10</td><td>0</td></tr><tr><td>50</td><td>0</td><td>0</td><td>80</td><td>10</td><td>45</td><td>10</td><td>0</td></tr></table> For example, the Tower '546 Accused Process comprises causing a portion of the first liner layer to diffuse into the upper portion of the semiconductor substrate, as shown in the image below. The diffused portion of the first liner layer forms a first intermix region at the upper portion of the substrate. The green arrow in the image below corresponds to the x-axis in the ensuing graph. That is, the upward direction in the image below corresponds to right-to-left in the ensuing graph.	Position (nm)	N (%)	O (%)	Si (%)	Ti (%)	Ni (%)	W (%)	Pt (%)	0	0	0	0	0	0	80	0	10	0	0	0	5	5	10	0	15	0	0	0	45	15	10	0	20	0	0	0	20	45	10	0	25	0	0	0	10	45	10	0	30	0	0	0	10	45	10	0	35	0	0	0	10	45	10	0	40	0	0	80	10	45	10	0	45	0	0	80	10	45	10	0	50	0	0	80	10	45	10	0
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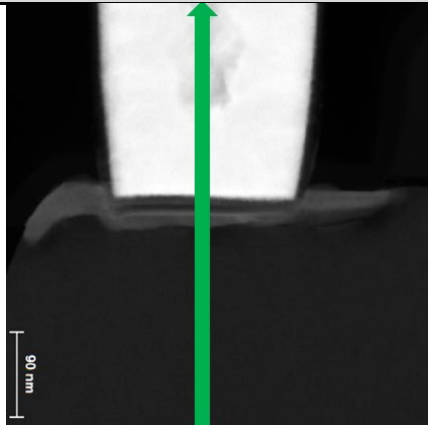
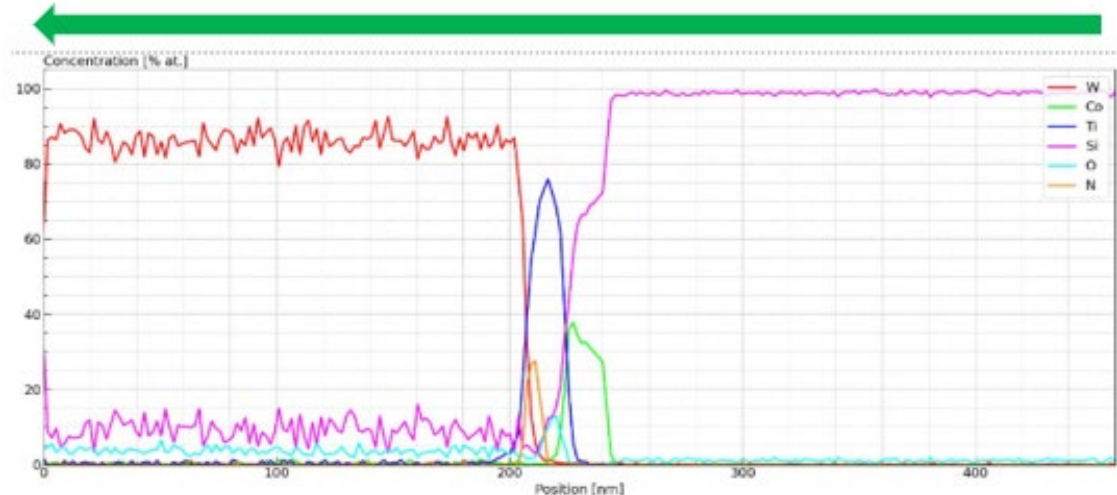
Claim	Tower '546 Accused Process																																																																																																																																																																								
	 As shown in the material analysis chart below, the intermix region created by the Tower '546 Accused Process includes a portion of the first liner layer diffusing into the silicon substrate. <table><caption>Approximate data from Material Analysis Chart</caption><thead><tr><th>Position [nm]</th><th>W [% at.]</th><th>Co [% at.]</th><th>Ti [% at.]</th><th>Si [% at.]</th><th>O [% at.]</th><th>N [% at.]</th></tr></thead><tbody><tr><td>0</td><td>85</td><td>0</td><td>0</td><td>10</td><td>5</td><td>0</td></tr><tr><td>100</td><td>85</td><td>0</td><td>0</td><td>10</td><td>5</td><td>0</td></tr><tr><td>200</td><td>85</td><td>0</td><td>0</td><td>10</td><td>5</td><td>0</td></tr><tr><td>210</td><td>85</td><td>30</td><td>75</td><td>65</td><td>10</td><td>25</td></tr><tr><td>220</td><td>85</td><td>35</td><td>75</td><td>65</td><td>10</td><td>25</td></tr><tr><td>230</td><td>85</td><td>30</td><td>75</td><td>65</td><td>10</td><td>25</td></tr><tr><td>240</td><td>85</td><td>30</td><td>75</td><td>65</td><td>10</td><td>25</td></tr><tr><td>250</td><td>85</td><td>30</td><td>75</td><td>65</td><td>10</td><td>25</td></tr><tr><td>260</td><td>85</td><td>30</td><td>75</td><td>65</td><td>10</td><td>25</td></tr><tr><td>270</td><td>85</td><td>30</td><td>75</td><td>65</td><td>10</td><td>25</td></tr><tr><td>280</td><td>85</td><td>30</td><td>75</td><td>65</td><td>10</td><td>25</td></tr><tr><td>290</td><td>85</td><td>30</td><td>75</td><td>65</td><td>10</td><td>25</td></tr><tr><td>300</td><td>85</td><td>30</td><td>75</td><td>65</td><td>10</td><td>25</td></tr><tr><td>310</td><td>85</td><td>30</td><td>75</td><td>65</td><td>10</td><td>25</td></tr><tr><td>320</td><td>85</td><td>30</td><td>75</td><td>65</td><td>10</td><td>25</td></tr><tr><td>330</td><td>85</td><td>30</td><td>75</td><td>65</td><td>10</td><td>25</td></tr><tr><td>340</td><td>85</td><td>30</td><td>75</td><td>65</td><td>10</td><td>25</td></tr><tr><td>350</td><td>85</td><td>30</td><td>75</td><td>65</td><td>10</td><td>25</td></tr><tr><td>360</td><td>85</td><td>30</td><td>75</td><td>65</td><td>10</td><td>25</td></tr><tr><td>370</td><td>85</td><td>30</td><td>75</td><td>65</td><td>10</td><td>25</td></tr><tr><td>380</td><td>85</td><td>30</td><td>75</td><td>65</td><td>10</td><td>25</td></tr><tr><td>390</td><td>85</td><td>30</td><td>75</td><td>65</td><td>10</td><td>25</td></tr><tr><td>400</td><td>85</td><td>30</td><td>75</td><td>65</td><td>10</td><td>25</td></tr></tbody></table> GF's contentions are based on information reasonably available at this stage of the case and GF's investigation to date. GF reserves the right to rely on reverse engineering and/or other testing, which will further confirm the Tower '546 Accused Process comprises causing a portion of the first liner layer to diffuse	Position [nm]	W [% at.]	Co [% at.]	Ti [% at.]	Si [% at.]	O [% at.]	N [% at.]	0	85	0	0	10	5	0	100	85	0	0	10	5	0	200	85	0	0	10	5	0	210	85	30	75	65	10	25	220	85	35	75	65	10	25	230	85	30	75	65	10	25	240	85	30	75	65	10	25	250	85	30	75	65	10	25	260	85	30	75	65	10	25	270	85	30	75	65	10	25	280	85	30	75	65	10	25	290	85	30	75	65	10	25	300	85	30	75	65	10	25	310	85	30	75	65	10	25	320	85	30	75	65	10	25	330	85	30	75	65	10	25	340	85	30	75	65	10	25	350	85	30	75	65	10	25	360	85	30	75	65	10	25	370	85	30	75	65	10	25	380	85	30	75	65	10	25	390	85	30	75	65	10	25	400	85	30	75	65	10	25
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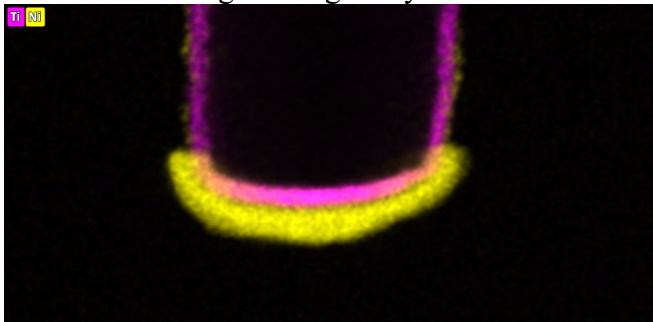
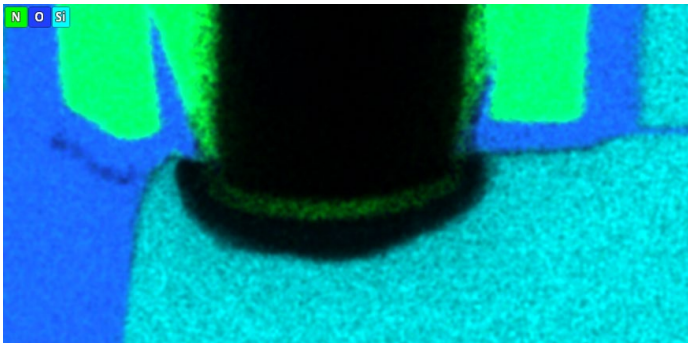
Claim	Tower '546 Accused Process
	into the upper portion of the semiconductor substrate to form a first intermix region at the upper portion of the semiconductor substrate. To the extent this limitation is found not to be literally present, this limitation is present under the doctrine of equivalents because the Tower '546 Accused Process performs substantially the same function, in substantially the same way, to achieve substantially the same result as the claimed limitation, and any differences are insubstantial. To date, Respondents have not alleged that this limitation is not literally present in the Tower '546 Accused Process.
[1d] depositing a refractory metal layer over the first intermix region; and	The Tower '546 Accused Process comprises depositing a refractory metal layer over the first intermix region. For example, Tower '546 Accused Process includes depositing a refractory metal layer over the first intermix region, as shown in the reverse engineering analysis below. An EDS line scan image showing a cross-section of a semiconductor device. The image displays two distinct peaks: a purple peak labeled 'Ti' (Titanium) and a yellow peak labeled 'Ni' (Nickel). The peaks are positioned at the base of a structure, indicating the presence of these elements in the intermix region. An EDS map image showing a cross-section of a semiconductor device. The image displays three distinct regions: a green region labeled 'N' (Nitrogen), a blue region labeled 'O' (Oxygen), and a cyan region labeled 'Si' (Silicon). The regions are positioned at the base of a structure, indicating the presence of these elements in the intermix region.

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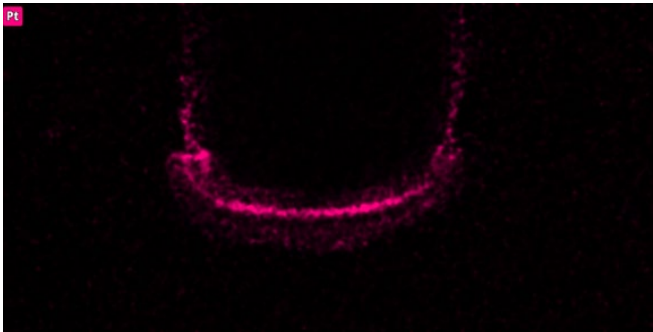
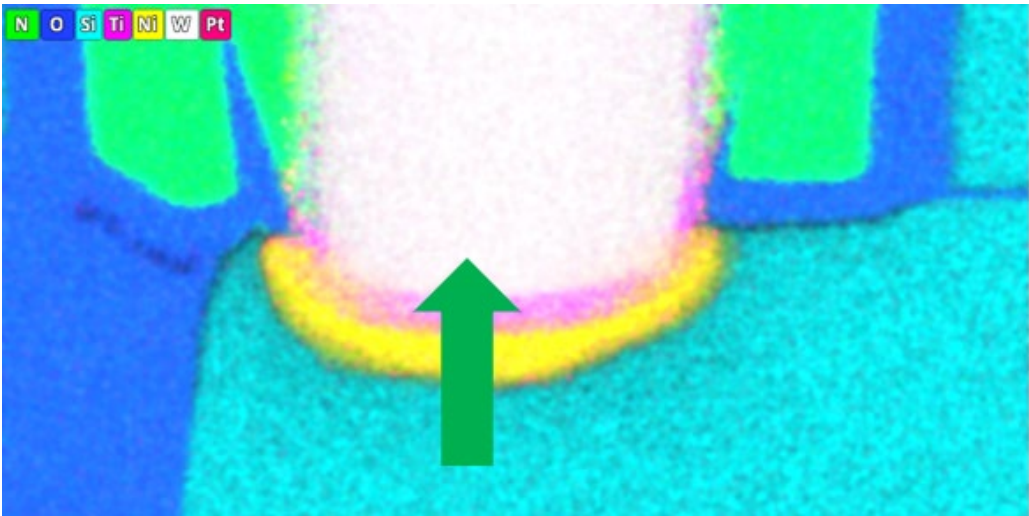
Claim	Tower '546 Accused Process
	 Again, the arrow in the reverse engineering analysis below corresponds to the arrow at the top of the ensuing material analysis chart.  As shown in the material analysis chart below, the refractory metal layer is deposited over the first intermix region.

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Claim	Tower '546 Accused Process
	Weight fraction (%) Position (nm) Legend: N (green), O (blue), Si (cyan), Ti (magenta), Ni (yellow), W (white), Pt (pink) For example, the Tower '546 Accused Process comprises depositing a refractory metal layer over the first intermix region.

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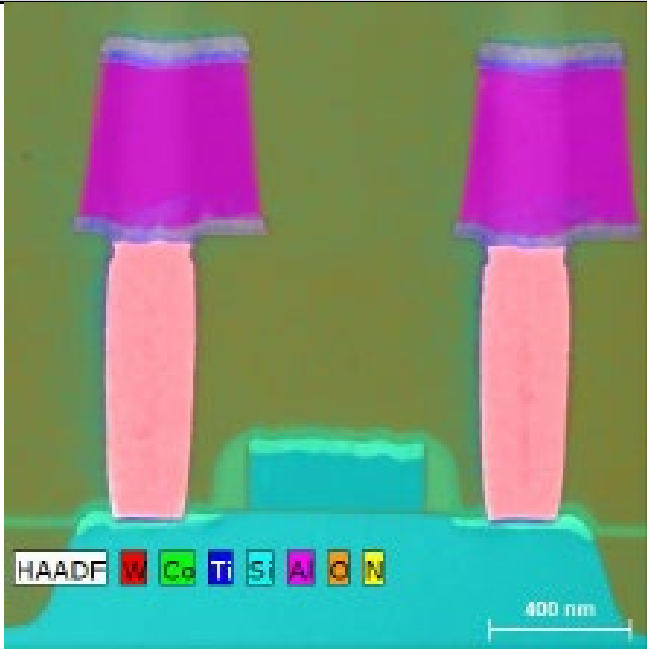
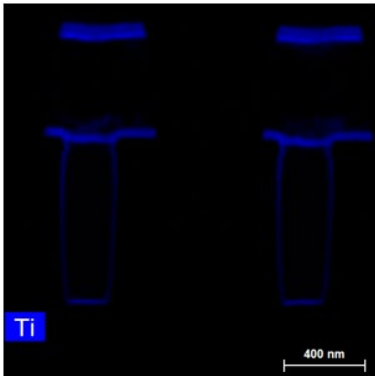
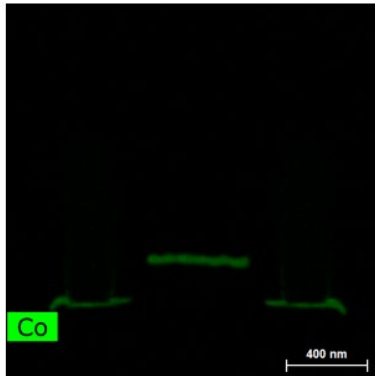
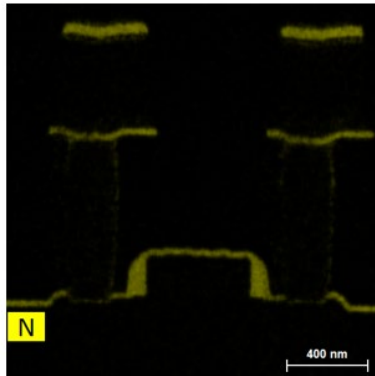
Claim	Tower '546 Accused Process
	    Again, the green arrow in the reverse engineering image below corresponds to the x-axis in the ensuing material analysis chart.

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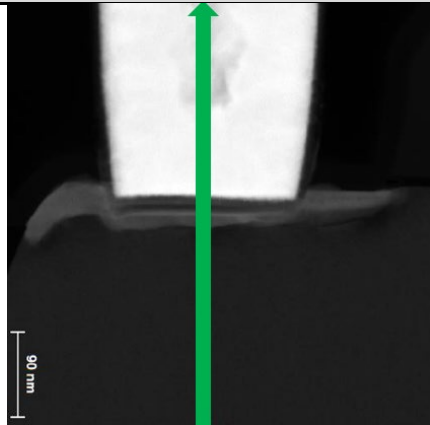
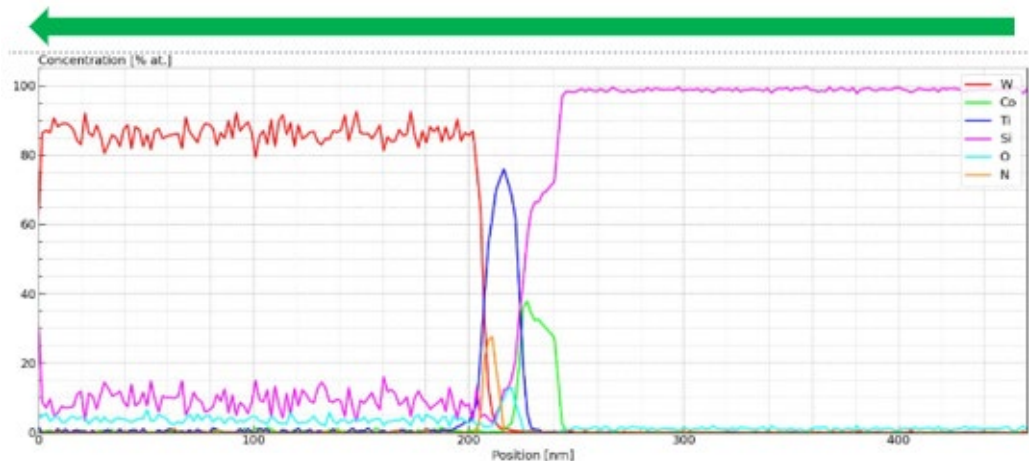
Claim	Tower '546 Accused Process																																																																																																																																																																																																																																														
	 As shown in the material analysis chart below, the refractory metal layer is deposited to the left of (i.e., on top of) the intermix region. <table><caption>Approximate data from Material Analysis Chart</caption><thead><tr><th>Position [nm]</th><th>W [% at.]</th><th>Co [% at.]</th><th>Ti [% at.]</th><th>Si [% at.]</th><th>O [% at.]</th><th>N [% at.]</th></tr></thead><tbody><tr><td>0</td><td>85</td><td>0</td><td>0</td><td>10</td><td>5</td><td>0</td></tr><tr><td>100</td><td>85</td><td>0</td><td>0</td><td>10</td><td>5</td><td>0</td></tr><tr><td>200</td><td>85</td><td>0</td><td>0</td><td>10</td><td>5</td><td>0</td></tr><tr><td>210</td><td>85</td><td>0</td><td>75</td><td>10</td><td>5</td><td>0</td></tr><tr><td>220</td><td>85</td><td>0</td><td>75</td><td>10</td><td>5</td><td>0</td></tr><tr><td>230</td><td>85</td><td>0</td><td>75</td><td>10</td><td>5</td><td>0</td></tr><tr><td>240</td><td>85</td><td>0</td><td>75</td><td>10</td><td>5</td><td>0</td></tr><tr><td>250</td><td>85</td><td>0</td><td>75</td><td>10</td><td>5</td><td>0</td></tr><tr><td>260</td><td>85</td><td>0</td><td>75</td><td>10</td><td>5</td><td>0</td></tr><tr><td>270</td><td>85</td><td>0</td><td>75</td><td>10</td><td>5</td><td>0</td></tr><tr><td>280</td><td>85</td><td>0</td><td>75</td><td>10</td><td>5</td><td>0</td></tr><tr><td>290</td><td>85</td><td>0</td><td>75</td><td>10</td><td>5</td><td>0</td></tr><tr><td>300</td><td>85</td><td>0</td><td>75</td><td>10</td><td>5</td><td>0</td></tr><tr><td>310</td><td>85</td><td>0</td><td>75</td><td>10</td><td>5</td><td>0</td></tr><tr><td>320</td><td>85</td><td>0</td><td>75</td><td>10</td><td>5</td><td>0</td></tr><tr><td>330</td><td>85</td><td>0</td><td>75</td><td>10</td><td>5</td><td>0</td></tr><tr><td>340</td><td>85</td><td>0</td><td>75</td><td>10</td><td>5</td><td>0</td></tr><tr><td>350</td><td>85</td><td>0</td><td>75</td><td>10</td><td>5</td><td>0</td></tr><tr><td>360</td><td>85</td><td>0</td><td>75</td><td>10</td><td>5</td><td>0</td></tr><tr><td>370</td><td>85</td><td>0</td><td>75</td><td>10</td><td>5</td><td>0</td></tr><tr><td>380</td><td>85</td><td>0</td><td>75</td><td>10</td><td>5</td><td>0</td></tr><tr><td>390</td><td>85</td><td>0</td><td>75</td><td>10</td><td>5</td><td>0</td></tr><tr><td>400</td><td>85</td><td>0</td><td>75</td><td>10</td><td>5</td><td>0</td></tr><tr><td>410</td><td>85</td><td>0</td><td>75</td><td>10</td><td>5</td><td>0</td></tr><tr><td>420</td><td>85</td><td>0</td><td>75</td><td>10</td><td>5</td><td>0</td></tr><tr><td>430</td><td>85</td><td>0</td><td>75</td><td>10</td><td>5</td><td>0</td></tr><tr><td>440</td><td>85</td><td>0</td><td>75</td><td>10</td><td>5</td><td>0</td></tr><tr><td>450</td><td>85</td><td>0</td><td>75</td><td>10</td><td>5</td><td>0</td></tr><tr><td>460</td><td>85</td><td>0</td><td>75</td><td>10</td><td>5</td><td>0</td></tr><tr><td>470</td><td>85</td><td>0</td><td>75</td><td>10</td><td>5</td><td>0</td></tr><tr><td>480</td><td>85</td><td>0</td><td>75</td><td>10</td><td>5</td><td>0</td></tr><tr><td>490</td><td>85</td><td>0</td><td>75</td><td>10</td><td>5</td><td>0</td></tr><tr><td>500</td><td>85</td><td>0</td><td>75</td><td>10</td><td>5</td><td>0</td></tr></tbody></table> GF's contentions are based on information reasonably available at this stage of the case and GF's investigation to date. GF reserves the right to rely on reverse engineering and/or other testing, which will further confirm the Tower '546 Accused Process comprises depositing a refractory metal layer over the first intermix region.	Position [nm]	W [% at.]	Co [% at.]	Ti [% at.]	Si [% at.]	O [% at.]	N [% at.]	0	85	0	0	10	5	0	100	85	0	0	10	5	0	200	85	0	0	10	5	0	210	85	0	75	10	5	0	220	85	0	75	10	5	0	230	85	0	75	10	5	0	240	85	0	75	10	5	0	250	85	0	75	10	5	0	260	85	0	75	10	5	0	270	85	0	75	10	5	0	280	85	0	75	10	5	0	290	85	0	75	10	5	0	300	85	0	75	10	5	0	310	85	0	75	10	5	0	320	85	0	75	10	5	0	330	85	0	75	10	5	0	340	85	0	75	10	5	0	350	85	0	75	10	5	0	360	85	0	75	10	5	0	370	85	0	75	10	5	0	380	85	0	75	10	5	0	390	85	0	75	10	5	0	400	85	0	75	10	5	0	410	85	0	75	10	5	0	420	85	0	75	10	5	0	430	85	0	75	10	5	0	440	85	0	75	10	5	0	450	85	0	75	10	5	0	460	85	0	75	10	5	0	470	85	0	75	10	5	0	480	85	0	75	10	5	0	490	85	0	75	10	5	0	500	85	0	75	10	5	0
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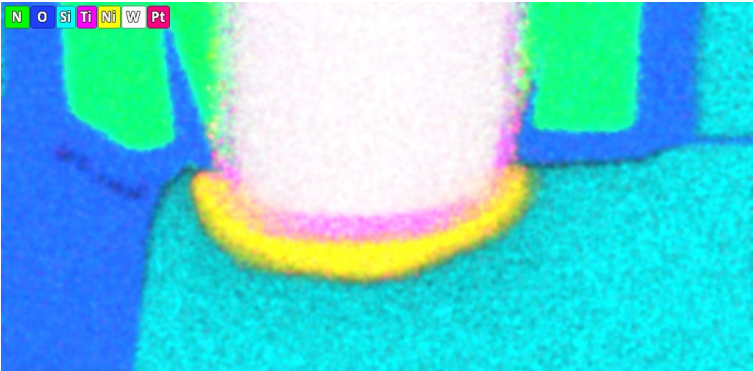
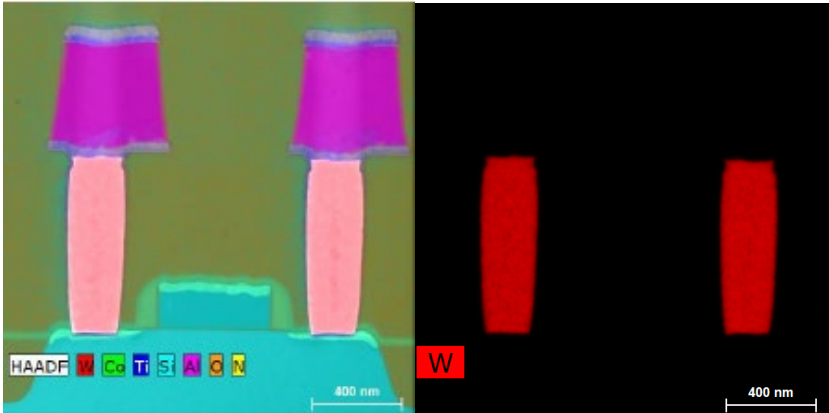
Claim	Tower '546 Accused Process
	To the extent this limitation is found not to be literally present, this limitation is present under the doctrine of equivalents because the Tower '546 Accused Process performs substantially the same function, in substantially the same way, to achieve substantially the same result as the claimed limitation, and any differences are insubstantial. To date, Respondents have not alleged that this limitation is not literally present in the Tower '546 Accused Process.
[1e] depositing a metal in the contact opening thereby forming the contact.	The Tower '546 Accused Process comprises depositing a metal in the contact opening thereby forming the contact.  

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Claim	Tower '546 Accused Process
	GF's contentions are based on information reasonably available at this stage of the case and GF's investigation to date. GF reserves the right to rely on reverse engineering and/or other testing, which will further confirm the Tower '546 Accused Process comprises depositing a metal in the contact opening thereby forming the contact. To the extent this limitation is found not to be literally present, this limitation is present under the doctrine of equivalents because the Tower '546 Accused Process performs substantially the same function, in substantially the same way, to achieve substantially the same result as the claimed limitation, and any differences are insubstantial. To date, Respondents have not alleged that this limitation is not literally present in the Tower '546 Accused Process.
[2pre] The method of claim 1, further comprising:	<i>See claim 1.</i>
[2a] removing a remaining non-diffused portion of the first liner layer to expose the first intermix region at the upper portion of the semiconductor substrate prior to depositing the refractory metal layer.	The Tower '546 Accused Process comprises removing a remaining non-diffused portion of the first liner layer to expose the first intermix region at the upper portion of the semiconductor substrate prior to depositing the refractory metal layer. GF's contentions are based on information reasonably available at this stage of the case and GF's investigation to date.

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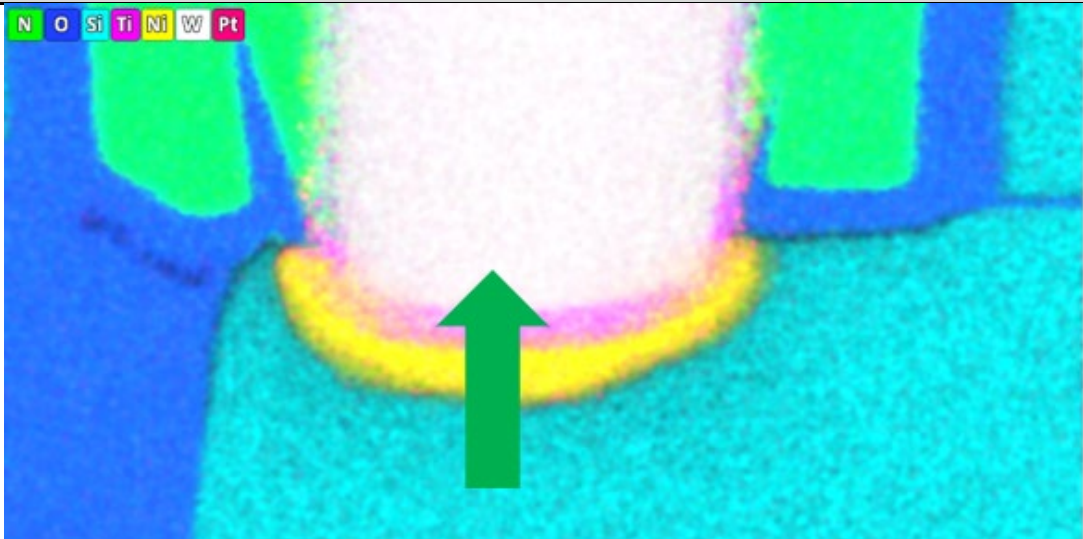
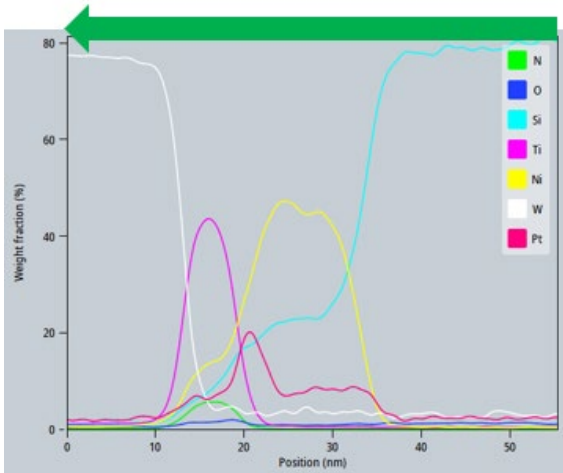
Claim	Tower '546 Accused Process																																																																																								
	 An EDS map showing the distribution of elements N, O, Si, Ti, Ni, W, and Pt. A green arrow points to a yellow region, which corresponds to Ni as indicated by the legend.  An EDS line scan graph showing the weight fraction (%) of elements N, O, Si, Ti, Ni, W, and Pt as a function of position (nm). A green arrow points to the Ni peak at approximately 25 nm. The graph shows a sharp peak for Ni (yellow) at approximately 25 nm, with smaller peaks for Ti (magenta) and W (white) at approximately 15 nm and 20 nm, respectively. The Si (cyan) signal increases significantly after 30 nm, reaching a plateau of about 80% weight fraction by 40 nm. The N (green) and O (blue) signals remain low throughout the scan. <table><tr><th>Position (nm)</th><th>N (%)</th><th>O (%)</th><th>Si (%)</th><th>Ti (%)</th><th>Ni (%)</th><th>W (%)</th><th>Pt (%)</th></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>10</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>15</td><td>0</td><td>0</td><td>0</td><td>10</td><td>0</td><td>10</td><td>0</td></tr><tr><td>20</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>10</td><td>0</td></tr><tr><td>25</td><td>0</td><td>0</td><td>0</td><td>0</td><td>45</td><td>0</td><td>0</td></tr><tr><td>30</td><td>0</td><td>0</td><td>0</td><td>0</td><td>45</td><td>0</td><td>0</td></tr><tr><td>35</td><td>0</td><td>0</td><td>20</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>40</td><td>0</td><td>0</td><td>80</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>45</td><td>0</td><td>0</td><td>80</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>50</td><td>0</td><td>0</td><td>80</td><td>0</td><td>0</td><td>0</td><td>0</td></tr></table> GF reserves the right to rely on reverse engineering and/or other testing, which will further confirm the Tower '546 Accused Process comprises removing a remaining non-diffused portion of the first liner layer	Position (nm)	N (%)	O (%)	Si (%)	Ti (%)	Ni (%)	W (%)	Pt (%)	0	0	0	0	0	0	0	0	10	0	0	0	0	0	0	0	15	0	0	0	10	0	10	0	20	0	0	0	0	0	10	0	25	0	0	0	0	45	0	0	30	0	0	0	0	45	0	0	35	0	0	20	0	0	0	0	40	0	0	80	0	0	0	0	45	0	0	80	0	0	0	0	50	0	0	80	0	0	0	0
Position (nm)	N (%)	O (%)	Si (%)	Ti (%)	Ni (%)	W (%)	Pt (%)																																																																																		
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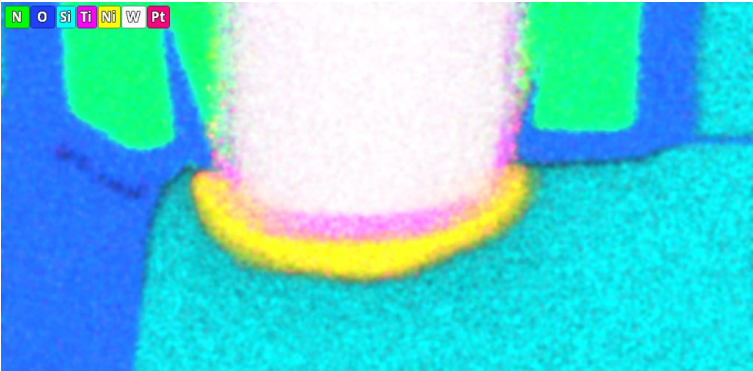
Claim	Tower '546 Accused Process
	to expose the first intermix region at the upper portion of the semiconductor substrate prior to depositing the refractory metal layer. To the extent this limitation is found not to be literally present, this limitation is present under the doctrine of equivalents because the Tower '546 Accused Process performs substantially the same function, in substantially the same way, to achieve substantially the same result as the claimed limitation, and any differences are insubstantial. To date, Respondents have not alleged that this limitation is not literally present in the Tower '546 Accused Process.
[3pre] The method of claim 1,	
[3a] wherein the depositing the first liner layer includes depositing at least one of: nickel platinum (NiPt), nickel (Ni), titanium (Ti), titanium silicide (TiSi), cobalt (Co), erbium (Er), platinum (Pt), and ytterbium (Yb).	The Tower '546 Accused Process comprises depositing a first liner layer that includes at least one of: nickel platinum (NiPt), nickel (Ni), titanium (Ti), titanium silicide (TiSi), cobalt (Co), erbium (Er), platinum (Pt), and ytterbium (Yb). GF's contentions are based on information reasonably available at this stage of the case and GF's investigation to date. 

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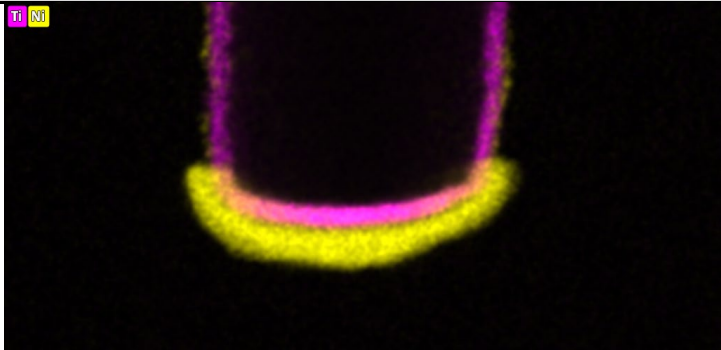
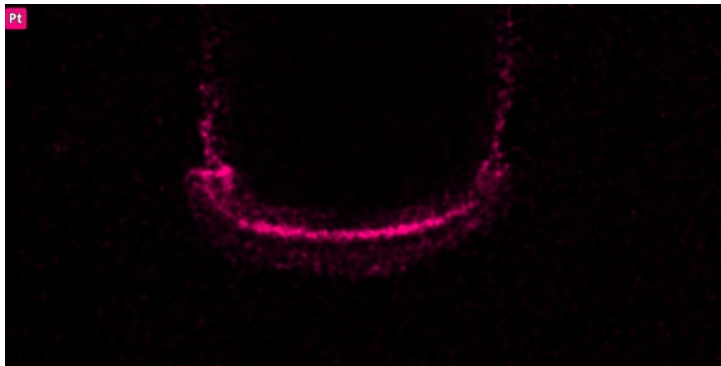
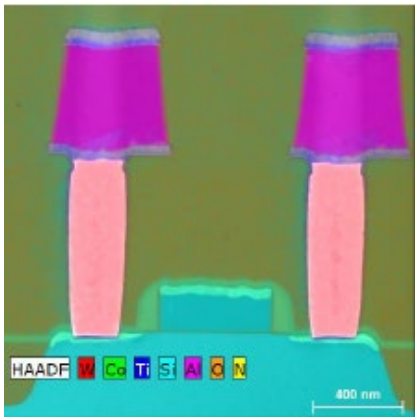
Claim	Tower '546 Accused Process
	 EDS map showing Ti (magenta) and Ni (yellow) distribution. The structure is U-shaped, with a thick yellow layer at the base and magenta layers on the sides and top.  EDS map showing Pt (magenta) distribution. The structure is U-shaped, with a thick magenta layer at the base and magenta layers on the sides and top.  HAADF image showing two pillars. The pillars are colored magenta (top) and pink (bottom). The base is blue. A scale bar indicates 400 nm. The legend at the bottom shows: HAADF, W, Co, Ti, Si, Al, O, N.

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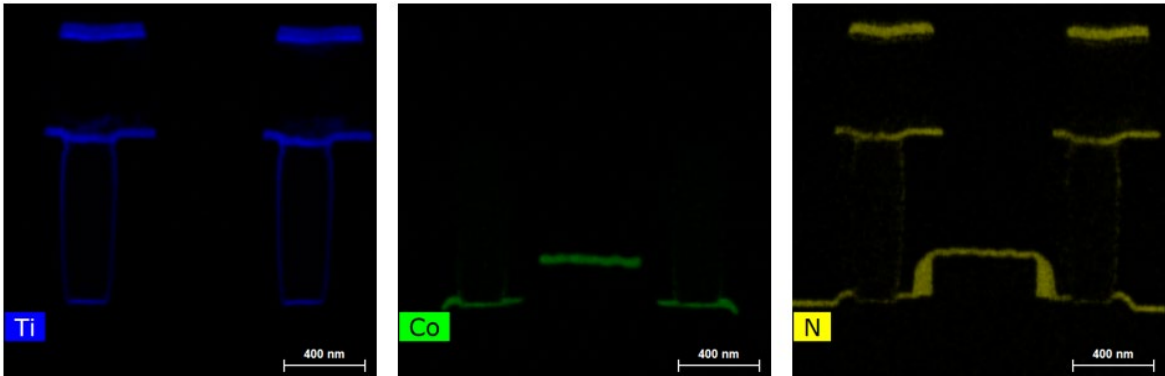
Claim	Tower '546 Accused Process
	 GF reserves the right to rely on reverse engineering and/or other testing, which will further confirm the Tower '546 Accused Process comprises depositing a first liner layer that includes at least one of: nickel platinum (NiPt), nickel (Ni), titanium (Ti), titanium silicide (TiSi), cobalt (Co), erbium (Er), platinum (Pt), and ytterbium (Yb). To the extent this limitation is found not to be literally present, this limitation is present under the doctrine of equivalents because the Tower '546 Accused Process performs substantially the same function, in substantially the same way, to achieve substantially the same result as the claimed limitation, and any differences are insubstantial. To date, Respondents have not alleged that this limitation is not literally present in the Tower '546 Accused Process.
[4pre] The method of claim 1,	See claim 1.
[4a] wherein depositing the refractory metal layer includes depositing a bilayer of titanium (Ti) and titanium nitride (TiN).	The Tower '546 Accused Process comprises depositing a refractory metal layer that includes a bilayer of titanium (Ti) and titanium nitride (TiN). GF's contentions are based on information reasonably available at this stage of the case and GF's investigation to date.

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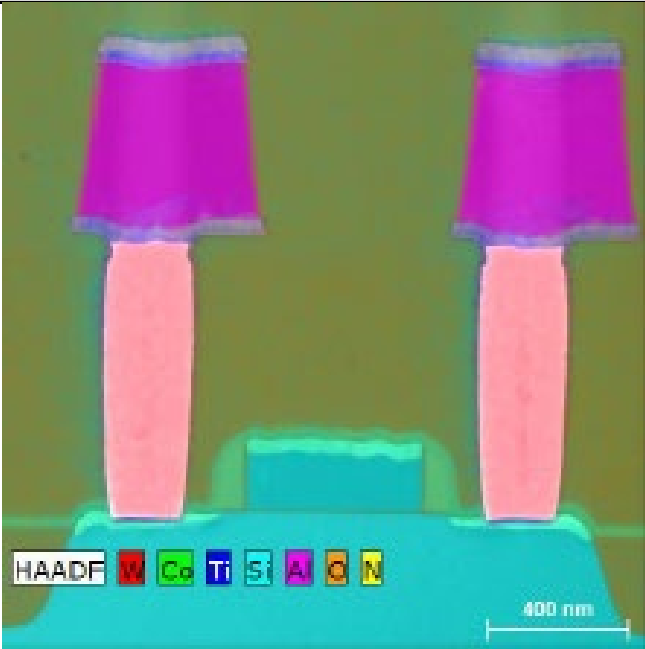
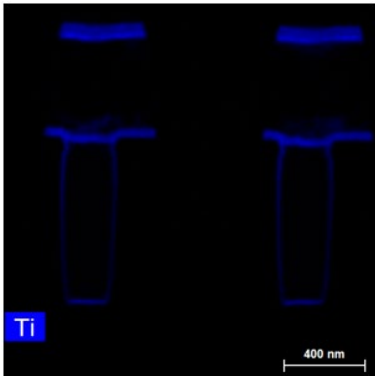
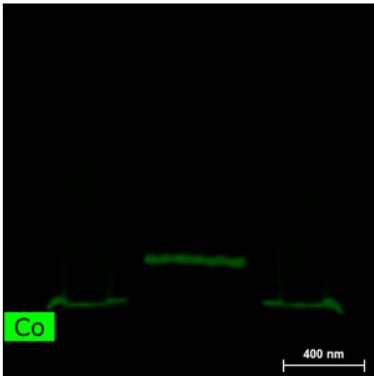
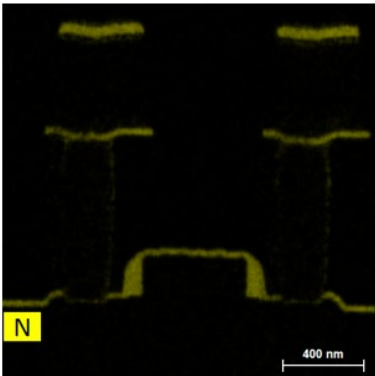
Claim	Tower '546 Accused Process
	   

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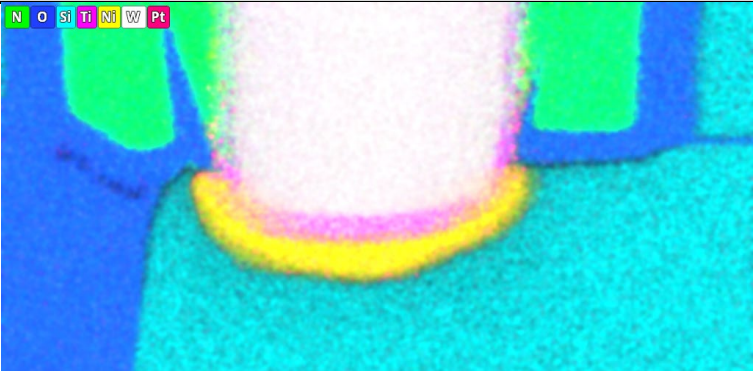
Claim	Tower '546 Accused Process
	 GF reserves the right to rely on reverse engineering and/or other testing, which will further confirm the Tower '546 Accused Process comprises depositing a refractory metal layer that includes a bilayer of titanium (Ti) and titanium nitride (TiN). To the extent this limitation is found not to be literally present, this limitation is present under the doctrine of equivalents because the Tower '546 Accused Process performs substantially the same function, in substantially the same way, to achieve substantially the same result as the claimed limitation, and any differences are insubstantial. To date, Respondents have not alleged that this limitation is not literally present in the Tower '546 Accused Process.
[5pre] The method of claim 1,	See claim 1.
[5a] wherein depositing the first liner layer includes performing physical vapor deposition.	In the Tower '546 Accused Process, depositing the first liner layer includes performing physical vapor deposition. GF's contentions are based on information reasonably available at this stage of the case and GF's investigation to date. See claim limitations [1b] and [3a]. GF reserves the right to rely on reverse engineering and/or other testing, which will further confirm that, in the Tower '546 Accused Process, depositing the first liner layer includes performing physical vapor deposition. To the extent this limitation is found not to be literally present, this limitation is present under the doctrine of equivalents because the Tower '546 Accused Process performs substantially the same function, in

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Claim	Tower '546 Accused Process
	substantially the same way, to achieve substantially the same result as the claimed limitation, and any differences are insubstantial. To date, Respondents have not alleged that this limitation is not literally present in the Tower '546 Accused Process.
[6pre] The method of claim 1, further comprising:	<i>See claim 1.</i>
[6a] polishing the metal; and	The Tower '546 Accused Process comprises polishing the metal. GF's contentions are based on information reasonably available at this stage of the case and GF's investigation to date. GF reserves the right to rely on reverse engineering and/or other testing, which will further confirm the Tower '546 Accused Process comprises polishing the metal. To the extent this limitation is found not to be literally present, this limitation is present under the doctrine of equivalents because the Tower '546 Accused Process performs substantially the same function, in substantially the same way, to achieve substantially the same result as the claimed limitation, and any differences are insubstantial. To date, Respondents have not alleged that this limitation is not literally present in the Tower '546 Accused Process.

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Claim	Tower '546 Accused Process
[6b] performing a laser anneal of the opening such that a second intermix region is formed at the upper portion of the semiconductor substrate and a bottom surface of the contact, wherein the second intermix region is composed of materials of the first intermix region and a material used for the refractory metal layer.	The Tower '546 Accused Process comprises performing a laser anneal of the opening such that a second intermix region is formed at the upper portion of the semiconductor substrate and a bottom surface of the contact, wherein the second intermix region is composed of materials of the first intermix region and a material used for the refractory metal layer. The graph displays the concentration profiles of various elements across a position from 0 to 400 nm. The y-axis represents Concentration [% at.] from 0 to 100. The x-axis represents Position [nm] from 0 to 400. The legend indicates the following elements: W (red), Co (green), Ti (blue), Si (purple), O (cyan), and N (orange). The W profile (red) is high (around 80-90%) from 0 to 200 nm and then drops to 0. The Co profile (green) shows a sharp peak around 200 nm. The Ti profile (blue) shows a sharp peak around 200 nm. The Si profile (purple) is low until 200 nm, then rises to about 80% and remains high. The O profile (cyan) is low throughout. The N profile (orange) shows a small peak around 200 nm.

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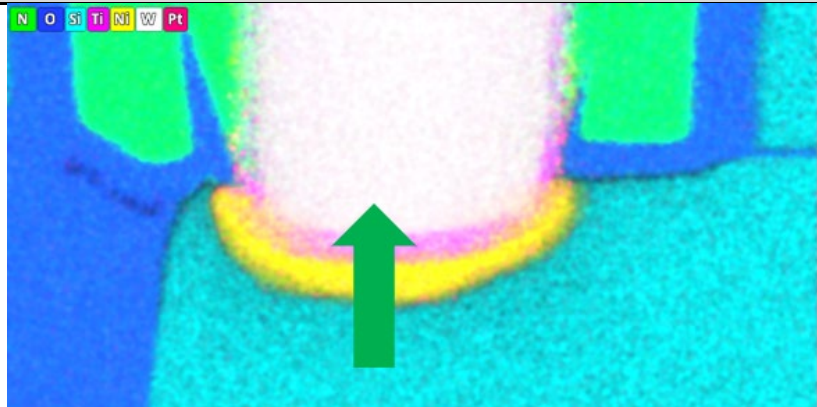
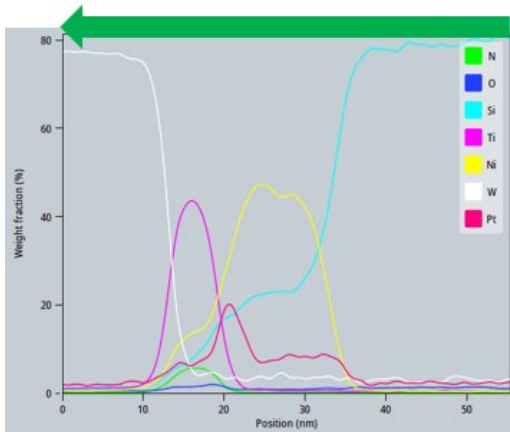
Claim	Tower '546 Accused Process																																																																																
	 <table><caption>Approximate data from EDS line scan graph</caption><tr><th>Position (nm)</th><th>N (%)</th><th>O (%)</th><th>Si (%)</th><th>Ti (%)</th><th>Ni (%)</th><th>W (%)</th><th>Pt (%)</th></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>10</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>75</td><td>0</td></tr><tr><td>15</td><td>0</td><td>0</td><td>0</td><td>45</td><td>10</td><td>20</td><td>0</td></tr><tr><td>20</td><td>0</td><td>0</td><td>0</td><td>20</td><td>45</td><td>10</td><td>0</td></tr><tr><td>25</td><td>0</td><td>0</td><td>0</td><td>10</td><td>45</td><td>10</td><td>0</td></tr><tr><td>30</td><td>0</td><td>0</td><td>0</td><td>5</td><td>45</td><td>10</td><td>0</td></tr><tr><td>35</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>75</td></tr><tr><td>40</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>75</td></tr><tr><td>50</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>75</td></tr></table> GF's contentions are based on information reasonably available at this stage of the case and GF's investigation to date. GF reserves the right to rely on reverse engineering and/or other testing, which will further confirm the Tower '546 Accused Process comprises performing a laser anneal of the opening such that a second intermix region is formed at the upper portion of the semiconductor substrate and a bottom surface of the contact, wherein the second intermix region is composed of materials of the first intermix region and a material used for the refractory metal layer.	Position (nm)	N (%)	O (%)	Si (%)	Ti (%)	Ni (%)	W (%)	Pt (%)	0	0	0	0	0	0	0	0	10	0	0	0	0	0	75	0	15	0	0	0	45	10	20	0	20	0	0	0	20	45	10	0	25	0	0	0	10	45	10	0	30	0	0	0	5	45	10	0	35	0	0	0	0	0	0	75	40	0	0	0	0	0	0	75	50	0	0	0	0	0	0	75
Position (nm)	N (%)	O (%)	Si (%)	Ti (%)	Ni (%)	W (%)	Pt (%)																																																																										
0	0	0	0	0	0	0	0																																																																										
10	0	0	0	0	0	75	0																																																																										
15	0	0	0	45	10	20	0																																																																										
20	0	0	0	20	45	10	0																																																																										
25	0	0	0	10	45	10	0																																																																										
30	0	0	0	5	45	10	0																																																																										
35	0	0	0	0	0	0	75																																																																										
40	0	0	0	0	0	0	75																																																																										
50	0	0	0	0	0	0	75																																																																										

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Claim	Tower '546 Accused Process
	To the extent this limitation is found not to be literally present, this limitation is present under the doctrine of equivalents because the Tower '546 Accused Process performs substantially the same function, in substantially the same way, to achieve substantially the same result as the claimed limitation, and any differences are insubstantial. To date, Respondents have not alleged that this limitation is not literally present in the Tower '546 Accused Process.
[7pre] The method of claim 1,	<i>See</i> claim 1.
[7] wherein the depositing the first liner layer includes depositing nickel platinum (NiPt), the depositing the refractory metal layer includes depositing a bilayer of titanium (Ti) and titanium nitride (TiN), and the depositing the metal in the opening includes depositing tungsten (W).	In the Tower '546 Accused Process, depositing the first liner layer includes depositing the first liner layer includes depositing nickel platinum (NiPt), the depositing the refractory metal layer includes depositing a bilayer of titanium (Ti) and titanium nitride (TiN), and the depositing the metal in the opening includes depositing tungsten (W). <i>See</i> claim limitations [1e], [3a], and [4a]. GF's contentions are based on information reasonably available at this stage of the case and GF's investigation to date. GF reserves the right to rely on reverse engineering and/or other testing, which will further confirm the Tower '546 Accused Process comprises depositing a first liner layer that includes nickel platinum (NiPt), depositing a refractory metal layer that includes a bilayer of titanium (Ti) and titanium nitride (TiN), and depositing a metal in the opening that includes tungsten (W). To the extent this limitation is found not to be literally present, this limitation is present under the doctrine of equivalents because the Tower '546 Accused Process performs substantially the same function, in substantially the same way, to achieve substantially the same result as the claimed limitation, and any differences are insubstantial. To date, Respondents have not alleged that this limitation is not literally present in the Tower '546 Accused Process.
[8pre] A method for forming a set of contacts in a dielectric layer to a field effect transistor, the field effect transistor	To the extent the preamble is limiting, the Tower '546 Accused Process comprises a method for forming a set of contacts in a dielectric layer to a field effect transistor, the field effect transistor including a source, a drain, and a gate between the source and the drain. <i>See, e.g.</i>, limitation [1pre]. The process used to create the source and drain in the Tower '546 Accused Process is the same. The transistors formed by the Tower '546 Accused Process include a gate between the source and the drain.

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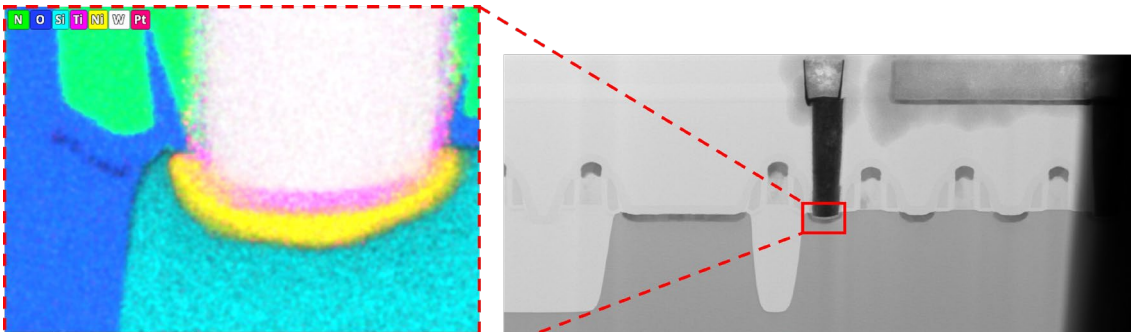
Claim	Tower '546 Accused Process
including a source, a drain, and a gate between the source and the drain, the method comprising:	GF's contentions are based on information reasonably available at this stage of the case and GF's investigation to date. GF reserves the right to rely on reverse engineering and/or other testing, which will further confirm the Tower '546 Accused Process comprises a method for forming a set of contacts in a dielectric layer to a field effect transistor, the field effect transistor including a source, a drain, and a gate between the source and the drain. To the extent this limitation is found not to be literally present, this limitation is present under the doctrine of equivalents because the Tower '546 Accused Process performs substantially the same function, in substantially the same way, to achieve substantially the same result as the claimed limitation, and any differences are insubstantial. To date, Respondents have not alleged that this limitation is not literally present in the Tower '546 Accused Process.
[8a] forming a contact opening in the dielectric layer to expose an upper-portion of each of the source and the drain;	The Tower '546 Accused Process comprises forming a contact opening in the dielectric layer to expose an upper-portion of each of the source and the drain. <i>See, e.g.</i>, limitation [1a]. 

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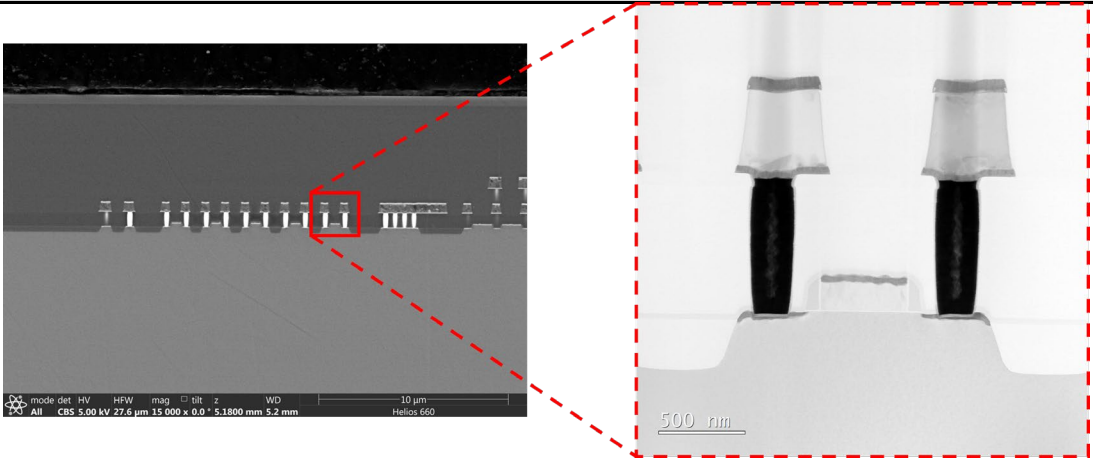
Claim	Tower '546 Accused Process
	 The image is a scanning electron micrograph (SEM) of a semiconductor device. A red box highlights a specific contact opening. A magnified inset, outlined with a red dashed line, shows a detailed view of this opening, revealing two vertical structures (source and drain) with a gap between them. A scale bar in the inset indicates 500 nm. The main image has a scale bar indicating 10 μm. Metadata at the bottom of the main image includes: mode det HV HPW mag tilt z WD 5.00 kV 27.6 μm 15 000 x 0.0° 5.1800 mm 5.2 mm Helios 660. GF's contentions are based on information reasonably available at this stage of the case and GF's investigation to date. GF reserves the right to rely on reverse engineering and/or other testing, which will further confirm the Tower '546 Accused Process comprises forming a contact opening in the dielectric layer to expose an upper-portion of each of the source and the drain. To the extent this limitation is found not to be literally present, this limitation is present under the doctrine of equivalents because the Tower '546 Accused Process performs substantially the same function, in substantially the same way, to achieve substantially the same result as the claimed limitation, and any differences are insubstantial. To date, Respondents have not alleged that this limitation is not literally present in the Tower '546 Accused Process.
[8b] depositing a first liner layer to conformally coat each contact opening and causing a portion of the first liner layer to diffuse into the upper portion of each of the source and the drain to form a first intermix region at the upper portion of the each of the source and the drain. <i>See, e.g.,</i> limitation [1b] and limitation [1c].	The Tower '546 Accused Process comprises depositing a first liner layer to conformally coat each contact opening and causing a portion of the first liner layer to diffuse into the upper portion of each of the source and the drain to form a first intermix region at the upper portion of the each of the source and the drain. <i>See, e.g.,</i> limitation [1b] and limitation [1c].

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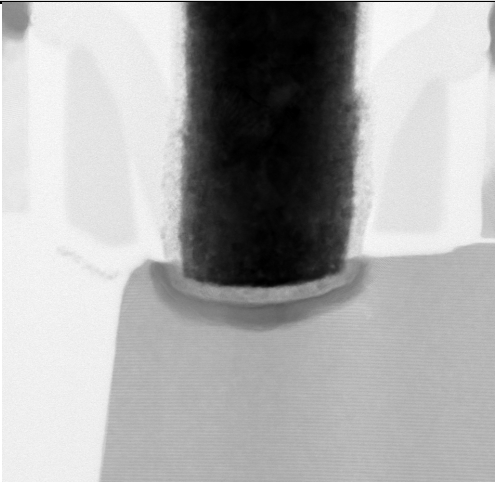
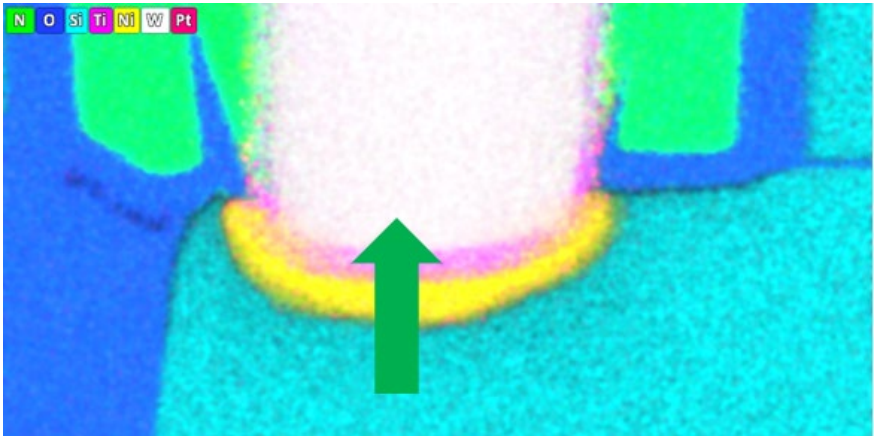
Claim	Tower '546 Accused Process
source and the drain to form a first intermix region at the upper portion of the each of the source and the drain;	Scanning electron micrograph (SEM) showing a cross-section of a device. A dark, vertical cylindrical feature is positioned on a light-colored, trapezoidal base. The interface between the cylinder and the base is visible. Energy-dispersive X-ray (EDS) elemental map of the same device structure. A legend at the top left identifies elements: N (green), O (blue), Si (light blue), Ti (pink), Ni (yellow), W (cyan), and Pt (magenta). The map shows a central light blue region (Si) surrounded by a magenta layer (Pt) and a yellow layer (Ni). A green arrow points to the interface between the yellow and magenta layers at the base of the central region.

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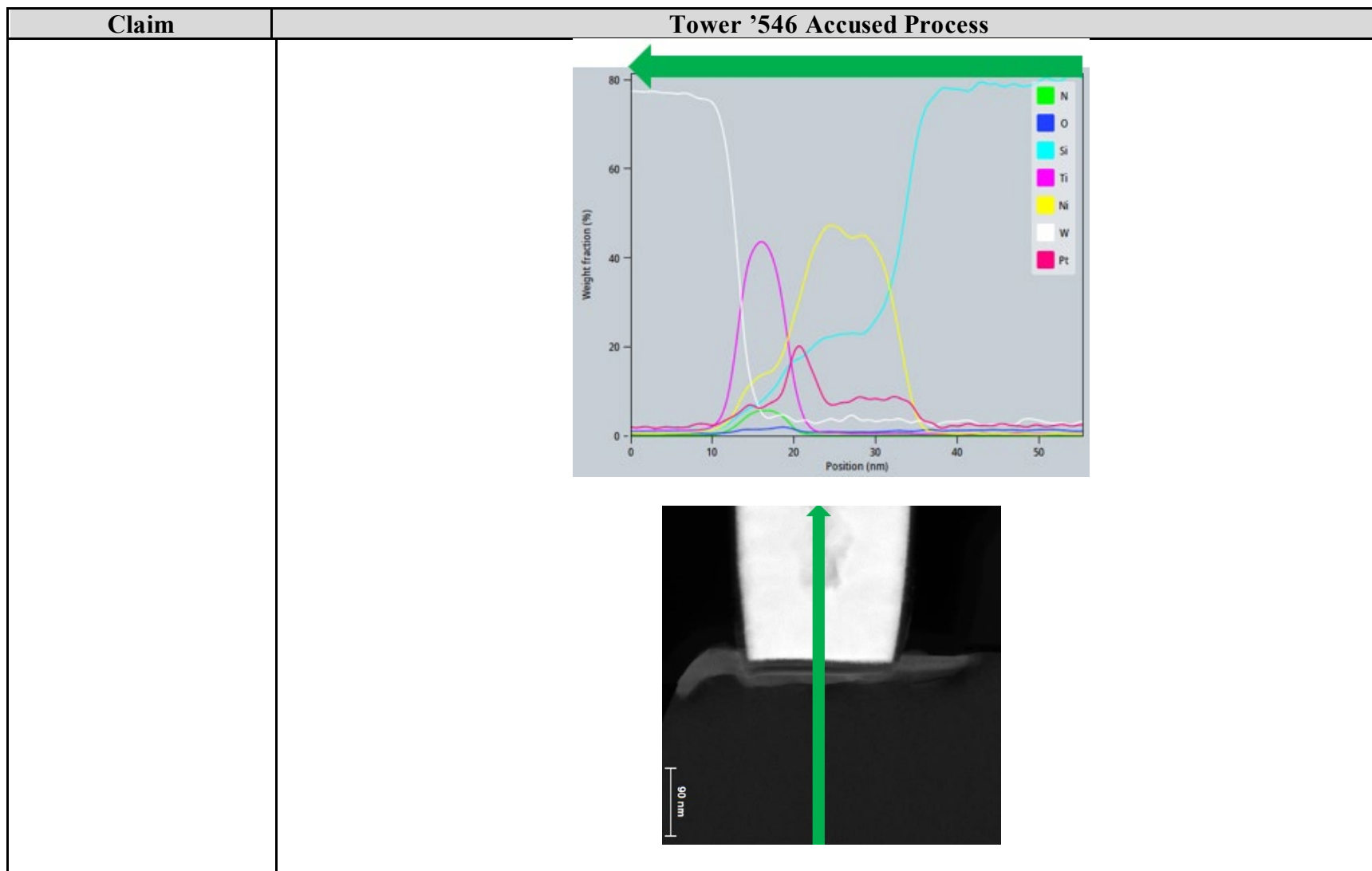


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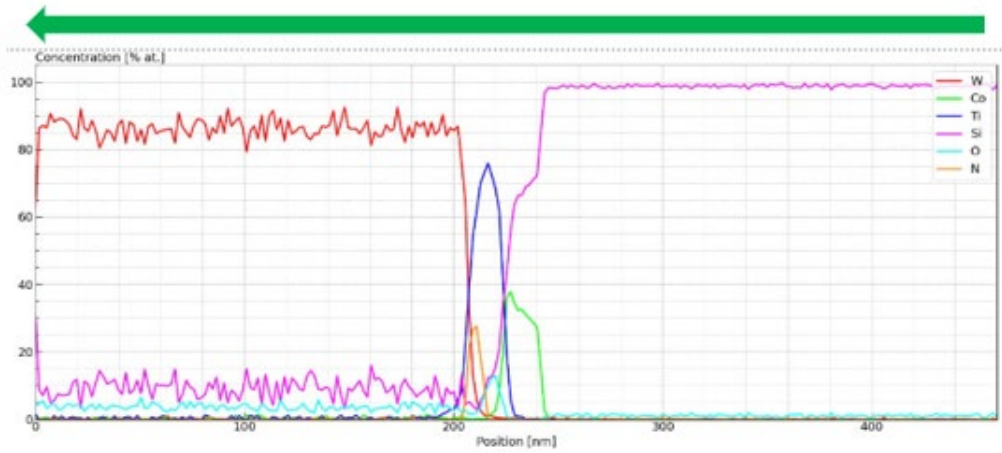
Claim	Tower '546 Accused Process
	 GF's contentions are based on information reasonably available at this stage of the case and GF's investigation to date. GF reserves the right to rely on reverse engineering and/or other testing, which will further confirm the Tower '546 Accused Process comprises depositing a first liner layer to conformally coat each contact opening and causing a portion of the first liner layer to diffuse into the upper portion of each of the source and the drain to form a first intermix region at the upper portion of each of the source and the drain. To the extent this limitation is found not to be literally present, this limitation is present under the doctrine of equivalents because the Tower '546 Accused Process performs substantially the same function, in substantially the same way, to achieve substantially the same result as the claimed limitation, and any differences are insubstantial. To date, Respondents have not alleged that this limitation is not literally present in the Tower '546 Accused Process.
[8c] depositing a refractory metal layer over the first intermix region; and	The Tower '546 Accused Process comprises depositing a refractory metal layer over the first intermix region. <i>See, e.g.,</i> limitation [1d].

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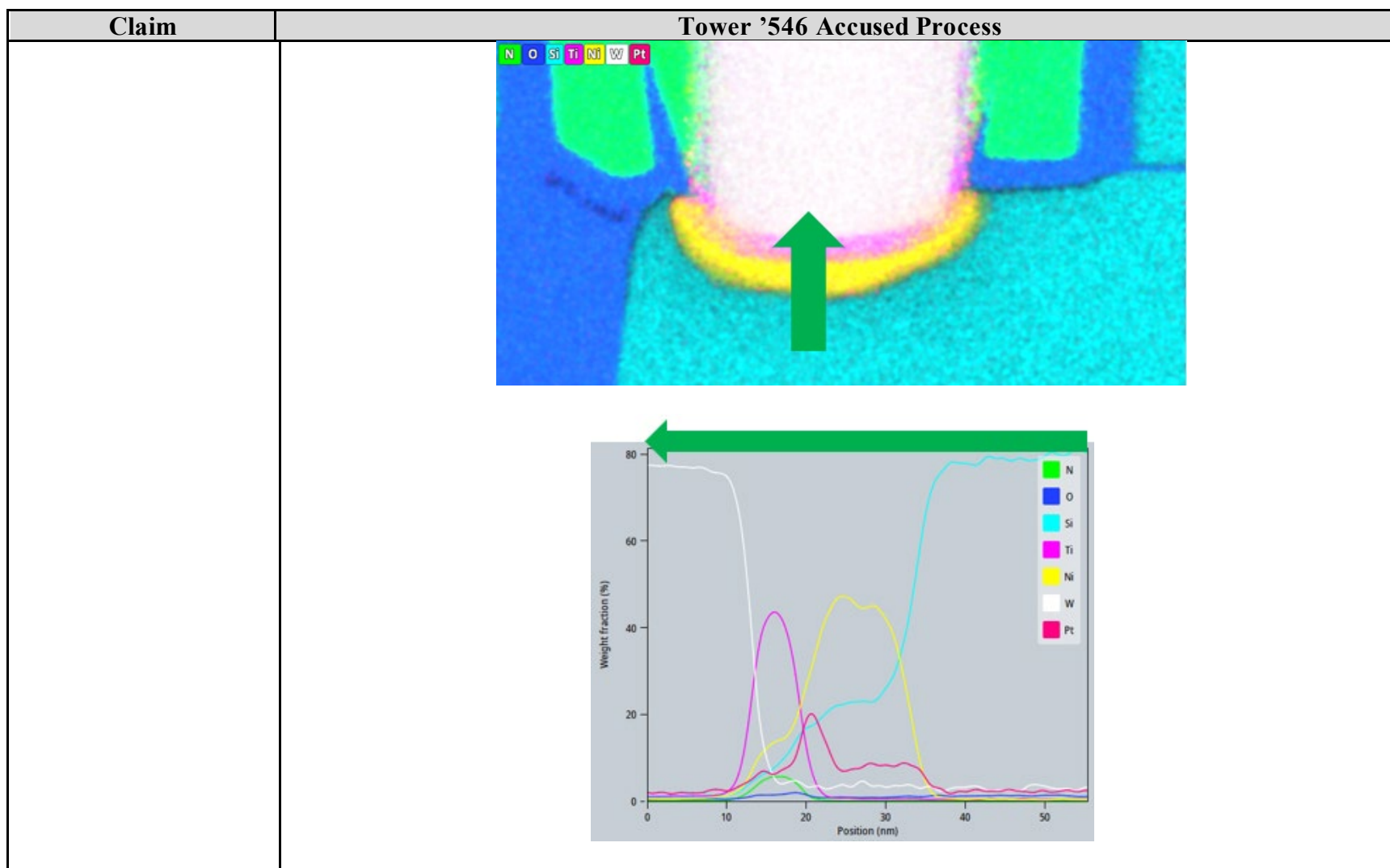


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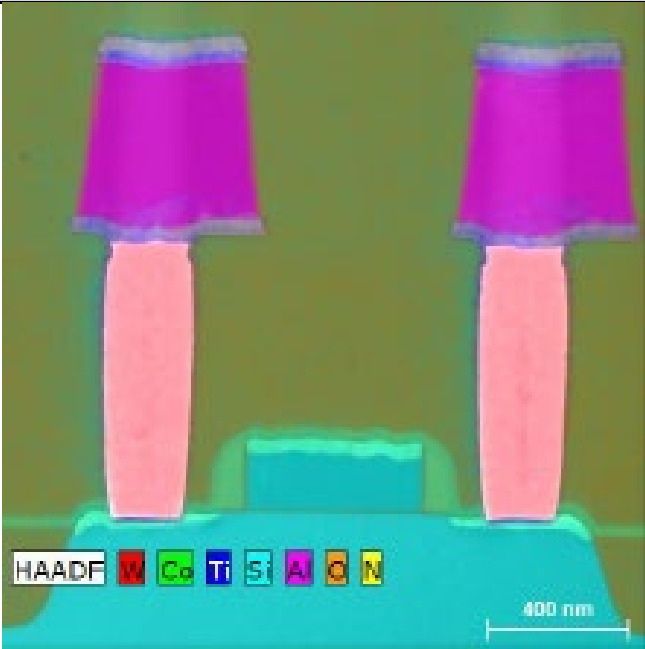
Claim	Tower '546 Accused Process
	 GF's contentions are based on information reasonably available at this stage of the case and GF's investigation to date. GF reserves the right to rely on reverse engineering and/or other testing, which will further confirm the Tower '546 Accused Process comprises depositing a refractory metal layer over the first intermix region. To the extent this limitation is found not to be literally present, this limitation is present under the doctrine of equivalents because the Tower '546 Accused Process performs substantially the same function, in substantially the same way, to achieve substantially the same result as the claimed limitation, and any differences are insubstantial. To date, Respondents have not alleged that this limitation is not literally present in the Tower '546 Accused Process.
[8d] depositing a metal to substantially coat the refractory metal layer and substantially fill	The Tower '546 Accused Process comprises depositing a metal to substantially coat the refractory metal layer and substantially fill each of the openings thereby forming the set of contacts. <i>See, e.g.</i> , limitation [1e].

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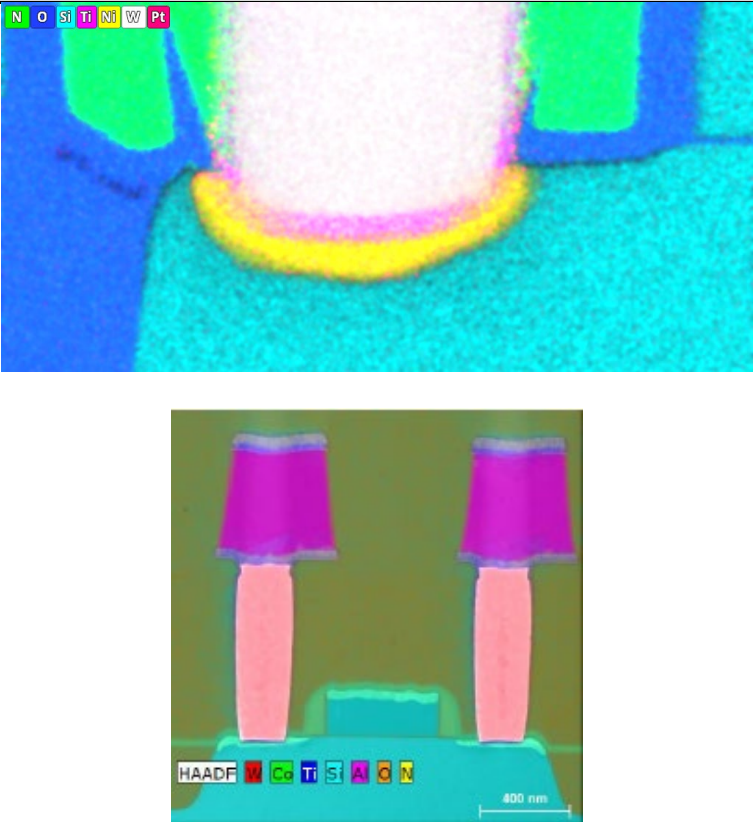
Claim	Tower '546 Accused Process
each of the openings thereby forming the set of contacts.	

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Claim	Tower '546 Accused Process
	GF's contentions are based on information reasonably available at this stage of the case and GF's investigation to date. GF reserves the right to rely on reverse engineering and/or other testing, which will further confirm the Tower '546 Accused Process comprises depositing a metal to substantially coat the refractory metal layer and substantially fill each of the openings thereby forming the set of contacts. To the extent this limitation is found not to be literally present, this limitation is present under the doctrine of equivalents because the Tower '546 Accused Process performs substantially the same function, in substantially the same way, to achieve substantially the same result as the claimed limitation, and any differences are insubstantial. To date, Respondents have not alleged that this limitation is not literally present in the Tower '546 Accused Process.
[9pre] The method of claim 8, further comprising:	See claim 8.
[9] removing a remaining non-diffused portion of the first liner layer to expose the intermix region prior to	The Tower '546 Accused Process comprises removing a remaining non-diffused portion of the first liner layer to expose the intermix region prior to depositing the refractory metal layer. See claim 2. GF's contentions are based on information reasonably available at this stage of the case and GF's investigation to date.

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Claim	Tower '546 Accused Process
depositing the refractory metal layer.	GF reserves the right to rely on reverse engineering and/or other testing, which will further confirm the Tower '546 Accused Process comprises removing a remaining non-diffused portion of the first liner layer to expose the intermix region prior to depositing the refractory metal layer. To the extent this limitation is found not to be literally present, this limitation is present under the doctrine of equivalents because the Tower '546 Accused Process performs substantially the same function, in substantially the same way, to achieve substantially the same result as the claimed limitation, and any differences are insubstantial. To date, Respondents have not alleged that this limitation is not literally present in the Tower '546 Accused Process.
[10pre] The method of claim 8,	<i>See</i> claim 8.
[10a] wherein the depositing the first liner layer includes depositing nickel platinum (NiPt), nickel (Ni), titanium (Ti), titanium silicide (TiSi), cobalt (Co), erbium (Er), platinum (Pt), and ytterbium (Yb).	The Tower '546 Accused Process comprises depositing a first liner layer that includes nickel platinum (NiPt), nickel (Ni), titanium (Ti), titanium silicide (TiSi), cobalt (Co), erbium (Er), platinum (Pt), or ytterbium (Yb). <i>See</i> claim 3. GF's contentions are based on information reasonably available at this stage of the case and GF's investigation to date. GF reserves the right to rely on reverse engineering and/or other testing, which will further confirm the Tower '546 Accused Process comprises depositing a first liner layer that includes nickel platinum (NiPt), nickel (Ni), titanium (Ti), titanium silicide (TiSi), cobalt (Co), erbium (Er), platinum (Pt), or ytterbium (Yb). To the extent this limitation is found not to be literally present, this limitation is present under the doctrine of equivalents because the Tower '546 Accused Process performs substantially the same function, in substantially the same way, to achieve substantially the same result as the claimed limitation, and any differences are insubstantial. To date, Respondents have not alleged that this limitation is not literally present in the Tower '546 Accused Process.
[11pre] The method of claim 8,	<i>See</i> claim 8.
[11a] wherein the depositing the	The Tower '546 Accused Process comprises depositing a refractory metal layer that includes a bilayer of titanium (Ti) and titanium nitride (TiN). <i>See</i> claim 4.

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Claim	Tower '546 Accused Process
refractory metal layer includes depositing a bilayer of titanium (Ti) and titanium nitride (TiN).	GF's contentions are based on information reasonably available at this stage of the case and GF's investigation to date. GF reserves the right to rely on reverse engineering and/or other testing, which will further confirm the Tower '546 Accused Process comprises depositing a refractory metal layer that includes a bilayer of titanium (Ti) and titanium nitride (TiN). To the extent this limitation is found not to be literally present, this limitation is present under the doctrine of equivalents because the Tower '546 Accused Process performs substantially the same function, in substantially the same way, to achieve substantially the same result as the claimed limitation, and any differences are insubstantial. To date, Respondents have not alleged that this limitation is not literally present in the Tower '546 Accused Process.
[12pre] The method of claim 8,	<i>See</i> claim 8.
[12a] wherein the depositing the first liner layer includes performing physical vapor deposition.	In the Tower '546 Accused Process, depositing the first liner layer comprises depositing the first liner layer using physical vapor deposition. <i>See</i> claim 5. GF's contentions are based on information reasonably available at this stage of the case and GF's investigation to date. GF reserves the right to rely on reverse engineering and/or other testing, which will further confirm the Tower '546 Accused Process comprises depositing the first liner layer using physical vapor deposition. To the extent this limitation is found not to be literally present, this limitation is present under the doctrine of equivalents because the Tower '546 Accused Process performs substantially the same function, in substantially the same way, to achieve substantially the same result as the claimed limitation, and any differences are insubstantial. To date, Respondents have not alleged that this limitation is not literally present in the Tower '546 Accused Process.
[13pre] The method of claim 8, further comprising:	<i>See</i> claim 8.

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Claim	Tower '546 Accused Process
[13a] polishing the metal in each of the openings; and	The Tower '546 Accused Process comprises polishing the metal in each of the openings. <i>See</i> claim limitation 6[a]. GF's contentions are based on information reasonably available at this stage of the case and GF's investigation to date. GF reserves the right to rely on reverse engineering and/or other testing, which will further confirm the Tower '546 Accused Process comprises polishing the metal in each of the openings. To the extent this limitation is found not to be literally present, this limitation is present under the doctrine of equivalents because the Tower '546 Accused Process performs substantially the same function, in substantially the same way, to achieve substantially the same result as the claimed limitation, and any differences are insubstantial. To date, Respondents have not alleged that this limitation is not literally present in the Tower '546 Accused Process.
[13b] performing a laser anneal of each of the openings such that a second intermix region is formed at the upper portion of the respective source and drain thereunder, wherein the second intermix region is composed of materials of the first intermix region and a material used for the refractory metal layer.	The Tower '546 Accused Process comprises performing a laser anneal of each of the openings such that a second intermix region is formed at the upper portion of the respective source and drain thereunder, wherein the second intermix region is composed of materials of the first intermix region and a material used for the refractory metal layer. <i>See</i> claim limitation [6b]. GF's contentions are based on information reasonably available at this stage of the case and GF's investigation to date. GF reserves the right to rely on reverse engineering and/or other testing, which will further confirm the Tower '546 Accused Process comprises performing a laser anneal of each of the openings such that a second intermix region is formed at the upper portion of the respective source and drain thereunder, wherein the second intermix region is composed of materials of the first intermix region and a material used for the refractory metal layer. To the extent this limitation is found not to be literally present, this limitation is present under the doctrine of equivalents because the Tower '546 Accused Process performs substantially the same function, in substantially the same way, to achieve substantially the same result as the claimed limitation, and any

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Claim	Tower '546 Accused Process
	differences are insubstantial. To date, Respondents have not alleged that this limitation is not literally present in the Tower '546 Accused Process.
[14pre] The method of claim 13,	<i>See</i> claim 13.
[14a] wherein the depositing the first liner layer includes depositing nickel platinum (NiPt), the depositing the refractory metal layer includes depositing a bilayer of titanium (Ti) and titanium nitride (TiN), and the depositing the metal in the opening includes depositing tungsten (W).	The Tower '546 Accused Process comprises depositing a first liner layer that includes nickel platinum (NiPt), depositing a refractory metal layer that includes a bilayer of titanium (Ti) and titanium nitride (TiN), and depositing a metal in the opening that includes tungsten (W). <i>See</i> claim 7. GF's contentions are based on information reasonably available at this stage of the case and GF's investigation to date. GF reserves the right to rely on reverse engineering and/or other testing, which will further confirm the Tower '546 Accused Process comprises depositing a first liner layer that includes nickel platinum (NiPt), depositing a refractory metal layer that includes a bilayer of titanium (Ti) and titanium nitride (TiN), and depositing a metal in the opening that includes tungsten (W). To the extent this limitation is found not to be literally present, this limitation is present under the doctrine of equivalents because the Tower '546 Accused Process performs substantially the same function, in substantially the same way, to achieve substantially the same result as the claimed limitation, and any differences are insubstantial. To date, Respondents have not alleged that this limitation is not literally present in the Tower '546 Accused Process.